Fingerprint Development

Reveal, record and compare fingerprints with three simple methods

AIM
To develop and record fingerprints on paper using different methods, to compare how well each method works, and to classify prints by their ridge patterns.

Background: why fingerprints?

The skin on our fingertips has tiny raised lines called friction ridges. When we touch a surface, sweat and natural oils on the ridges leave a copy of the pattern. These marks are usually invisible (latent prints) and can be made visible by chemicals or powders. Ridge patterns form before birth and stay the same through life, and no two people have ever been found to have identical prints — even identical twins differ — so fingerprints are used in forensic science and identity checks.

Pattern	Look	Roughly how common
Loop	Ridges enter on one side, curve and leave on the same side	about 60–65 %
Whorl	Circular or spiral ridges	about 30–35 %
Arch	Ridges rise in a wave across the finger	about 5 %

Supplies and equipment

For all methods	Iodine method (teacher)	Graphite or ink method
tweezers disposable gloves scissors 3 white index cards (3 × 5 in) and 3 pieces of white paper (3 × 5 in) per team magnifying glass, camera or phone	iodine crystals (about ½ teaspoon) TLC tank with lid blank glass TLC plate sticky tape glass tray and hot water	soft pencil (6B) or ink pad plain paper clear sticky tape white card

Preparing the fingerprint samples

1. Wear gloves when handling the items. Everyone first washes their hands, then waits until they are dry.
2. In dry weather, **improve the prints** by wiping your fingers on a sweaty part of your body (forehead or the side of the nose), or by rubbing in a little moisturising lotion.

3. Each group prepares **at least three fingerprints** on index cards and **three** on white paper. Press the finger firmly and evenly, hold for one second, and lift straight up. Do not slide the finger.
4. Label each card (name, finger, date). Keep a stock of cards for all the methods below.

Recording your results

- **Photograph** each print as soon as it is visible.
- **Write down:** (1) conditions — time, temperature, surface; (2) observations — how fast the print appeared, its colour and quality, problems and what you tried to fix them; (3) discussion — which method worked best, and how the surface (card or paper) mattered.

Method 1: Pencil-graphite and tape (safe for everyone)

1. Scribble thickly with a soft pencil on a piece of paper to make a dark patch.
2. Rub your finger on the patch, then press it on a clean part of a piece of clear tape stuck to a table. (Or press your finger on the graphite, then onto the tape.)
3. Peel the tape and stick it on a white card. The ridges show in grey.
4. Look at the print with the magnifier and decide: loop, whorl or arch?

Method 2: Ink-pad print (safe for everyone)

1. Roll one finger gently from one side to the other on a stamp pad.
2. Roll the finger once on a white card, using even pressure.
3. Compare it with the graphite print of the same finger. Which shows more detail?

Method 3: Iodine fuming (teacher only)

SAFETY — IODINE VAPOUR IS HARMFUL

- The **teacher** prepares and handles the iodine. Students observe from a distance.
- Work in a **well-ventilated room or a fume hood**. Wear gloves and goggles. Do not breathe the vapour.
- Iodine stains skin, clothes and surfaces brown. Keep the lid closed except when needed.
- Do not heat iodine directly. Only warm the tank in a tray of hot (not boiling) water.

How it works: iodine vapour dissolves in the oils left by the fingerprint, staining the ridges brown. The image is not permanent and fades in minutes, so photograph it as soon as it can be read.

1. Make an iodine fuming chamber: put about ½ teaspoon of iodine crystals on the bottom of a TLC development tank and close the lid.

2. Tape your fingerprint samples on a blank glass TLC plate, print side facing outwards.
3. Slide the plate into the tank so that the samples face the bottom of the tank (towards the iodine), and replace the lid.
4. Place the tank in a shallow tray of hot water to speed up the evaporation of iodine.
5. Watch through the glass. Prints begin to appear as brown ridges within a few minutes.
6. As soon as the print is clear, lift the plate with tweezers and **photograph it immediately**.
7. Close the tank, let it cool, and keep it sealed. Wash hands afterwards.

Analysis

Method	Time to see the print	Clarity (1–5)	On card vs paper	Problems
Graphite and tape				
Ink pad				
Iodine fuming				

Questions

1. Which pattern (loop, whorl, arch) is your commonest? Count your class and make a bar graph.
2. Which method gave the clearest print? Which was fastest? Which was the safest?
3. Why does a dry-skinned finger leave a weaker print?
4. Why can't you rely on a single partial print to identify someone? What else would a forensic scientist do?